

Adarsh Tripathi

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Education

Vellore Institute of Technology, Vellore	2023 – 2027
Bachelor of Technology in Computer Science Engineering	CGPA: 8.96
DAV Public School, Sreshtha Vihar, Delhi	2023
Senior Secondary (Class XII)	94%
DAV Public School, Sreshtha Vihar, Delhi	2021
Secondary (Class X)	88%

Skills

Programming: Python, Java, C, C++

Web Development: HTML, CSS, JavaScript, TypeScript, React, Node.js, Flask, Django, Express

Databases & Backend: MySQL, MongoDB, REST APIs, JWT

Tools: Git/GitHub, Linux, Docker, Kubernetes, Vertex AI, Google Colab, Power BI, Tableau, Excel

Libraries: NumPy, Pandas, Matplotlib, TensorFlow, Keras

CS Fundamentals: OOPS, DBMS, SQL, DSA, OS, CN, COA, System Design

Soft Skills: Ownership mindset, Problem Solving, Time Management, Adaptability

Projects

College RAG System (Python, LangChain, FAISS, Flask, LLM) *GitHub*
Built a production-grade Retrieval-Augmented Generation system serving 40,000+ VIT students, enabling semantic vector search over institutional knowledge bases for academic workflows, course registration, and administrative queries via context-aware LLM responses.

Zero-Trust SOC Platform (Python, Kafka, Docker, OpenSearch, Neo4j, Suricata, Wazuh) *GitHub*
Built a full-stack Security Operations Center on Zero-Trust principles with Kafka-driven telemetry ingestion, GNN-based UEBA, Sigma rule detection, and LLM-powered alert triage. Automated closed-loop SOAR responses with millisecond IP isolation, Keycloak SSO, and Hashicorp Vault secret management.

WaferMap Pattern Intelligence (Python, PyTorch/TensorFlow, CNNs, Computer Vision) *GitHub*
Deep Learning pipeline for semiconductor defect detection using CNNs and Autoencoders, achieving 99%+ accuracy in identifying spatial failure patterns and optimizing manufacturing yield.

Shiksha Sahayak (Node.js, MongoDB, Electron, REST APIs) *GitHub*
Offline-first education management suite with 10+ automated modules, reducing administrative overhead by 25% via local-to-cloud reconciliation.

Command Line Interface (V8 JavaScript Runtime, LLM Architectures, DevOps Tooling) *GitHub*
Unified CLI suite with 40+ developer utilities including LLM-driven code generation and SSH provisioning, reducing administrative latency by 30%.

Research & Publications

Patent Filed – Early Crop Disease Detection System *VIT, 2026*
Co-invented and filed a patent integrating Red-Edge/SWIR multispectral sensing, a cross-phenological Vision Transformer, and a graph-diffusion propagation engine for disease detection 7–10 days before visible symptoms.

Journal Paper (Under Review) – Quantum-Enhanced Crop Disease Detection
Proposed a Quantum-Enhanced Edge-AI framework for pre-symptomatic crop disease detection using temporal spectral intelligence, Red-Edge/SWIR sensor fusion, and quantum-inspired optimization. System architecture and experimental validation complete; manuscript under journal review.

Hackathons & Competitions

- **Deloitte HackSplosion 2026** – Built an AI-based solution as part of a team and advanced to the semi-final round of Deloitte's national GenW.AI hackathon.
- **Smart India Hackathon (SIH) 2025** – Cleared VIT's internal evaluation round and received a national-level shortlist among competing college teams.
- **BrAINWARS 2026** – Qualified through the Brain Blitz round of Bain Capability Network's national business case competition.
- **JPMorgan Chase Code for Good 2026** – Cleared the coding assessment and interview rounds for JPMC's flagship social-impact hackathon.

Certifications

• IBM Full Stack Software Developer Professional Certificate	<i>Certificate</i>
• Programming with Java – Amazon	<i>Certificate</i>
• Machine Learning – DeepLearning.AI (Stanford)	<i>Certificate</i>
• OCI Generative AI Professional Certificate – Oracle	<i>Certificate</i>
• Google Cybersecurity Professional Certificate	<i>Certificate</i>

Experience

IEEE Engineering in Medicine and Biology Society (EMBS), VIT	May 2024 – Dec 2024
• Organized technical workshops on biomedical engineering and healthcare technologies; coordinated interdisciplinary sessions and knowledge-sharing initiatives covering emerging trends in biotech and medical devices.	